

Modeling Water Management Dynamics in the Nile River Basin: Integrating Hydrological and Socioeconomic Factors for Sustainable Policy Analysis

Muhammad Ahmad

*Department of Electrical Engineering
Lahore University of Management Sciences*

Lahore, Pakistan
25100076@lums.edu.pk

Abstract—The Nile River Basin, a lifeline for over 250 million people across 11 countries, has been a major source of conflict in North-Eastern Africa and the Horn of Africa, intensified by competing demands for hydropower, irrigation, and domestic use. Historically, Ethiopia has been disproportionately unfavored in water distribution treaties despite being the source of the Blue Nile. The recently built and filled Grand Ethiopian Renaissance Dam (GERD) marks a significant shift in regional hydropolitics, amplifying tensions over water rights and directly threatening downstream water security amid rising socioeconomic and political pressures in Egypt and Sudan. This study explores how these pressures shape sustainable water allocation policies once GERD has completed its filling stages. We employ a system dynamics approach to capture feedback-driven dynamics unavailable in static models, and develop two interlinked models: (1) a calibrated hydrological model representing real-world flow dynamics, including GERD's filling, and (2) a flight-simulator model that overlays domestic, agricultural, energy, and diplomatic pressures. The models simulate the operation of GERD, the Aswan High Dam, and the Jebel Aulia Reservoir under safe capacity constraints. Continuing pre-GERD policies during filling results in a decline in either the storage or the releases of downstream reservoirs; Jebel Aulia requires a minimum operational efficiency of about 0.85 to avoid overtopping; and while constant moderate pressures can be sustained, high or linearly growing pressures, driven by agricultural and domestic demand, lead to infeasible operations. These findings underscore the need for coordinated reservoir operations and pressure limits in shaping transboundary water agreements.

Index Terms—system dynamics, transboundary water management, Nile River Basin, GERD, reservoir operations, water policy

I. INTRODUCTION

Water is vital for human survival, economic prosperity, and environmental health, yet its equitable distribution remains one of the world's most pressing challenges. Around four billion people experience severe physical water scarcity for at least one month per year, a figure projected to rise with population growth and climate change [1]. Transboundary river

basins support 40% of the global population and have been a primary source of contention over irrigation, hydropower, and domestic needs [2]. Such shared waters demand cooperative management, but historical agreements and unilateral infrastructure projects often exacerbate tensions, threatening regional stability and sustainable development.

The Nile River Basin, stretching across 11 countries and sustaining over 250 million people [3], is a stark example. Originating in Ethiopia's highlands via the Blue Nile and in the equatorial lakes via the White Nile, the river powers agriculture, industry, and livelihoods across the region. Egypt is nearly entirely dependent on the Nile for its water supply; Sudan relies on it for irrigation and hydropower; and Ethiopia seeks to harness its hydroelectric potential to drive economic growth. The 1959 Nile Waters Agreement allocated 55.5 billion cubic meters (BCM) annually to Egypt and 18.5 BCM to Sudan, ignoring Ethiopia, whose highlands supply about 85% of the Nile's flow through the Blue Nile and its eastern tributaries [3]. This exclusion fueled resentment and, ultimately, Ethiopia's construction of GERD, operational since 2020. With a capacity of 74 BCM, GERD has reshaped the basin's hydrology, reducing downstream flows to Egypt and Sudan during its filling phases and raising fears of water shortages.

The stakes are high. Egypt, with a population exceeding 100 million, faces existential risks if inflows to the Aswan High Dam (AHD) decline, threatening food security and urban water supplies. Sudan balances irrigation needs with hydropower goals and grapples with sediment and flow disruptions from upstream dams. Ethiopia views GERD as a cornerstone of its energy security, yet its operations strain diplomatic relations with Egypt and Sudan, which invoke historical rights to the Nile's waters. Rising populations, expanding agriculture, and growing energy demands further complicate allocation, while political pressures, such as Egypt's diplomatic leverage or Ethiopia's sovereignty claims, shape policy decisions. Climate variability, with Blue Nile inflows ranging from 0.68 to 13.5 BCM per month, adds urgency [3].

This study examines water allocation in the Nile Basin through a systems approach, focusing on Ethiopia, Sudan, and Egypt. We provide two models: a baseline physical water-balance model, calibrated with real flows and capacities and configured as a flight simulator for scenario testing; and a second model that builds on it by integrating dynamic socioeconomic and political pressures (domestic needs, agricultural demands, energy goals, and diplomatic negotiation) that hydrological models often overlook. We ask: *How do varying socioeconomic and political pressures influence sustainable water allocation policies in the Nile River Basin under steady-state dam operations?* The first model establishes the hydrological core, including GERD's filling; the second addresses the question in the post-filling steady state, under safe capacity constraints for each reservoir.

II. RELATED WORK

GERD's construction has led to numerous studies of its effects, particularly on the downstream countries, covering water allocation, dam operations, conflict resolution, and impacts on agriculture, food security, economy, and energy. Basheer et al. [4] investigate the water–energy–food nexus in the Blue Nile Basin to evaluate trade-offs in cooperative water management, quantifying GERD's impacts on hydropower, irrigation, and food security and highlighting Ethiopia's energy gains versus Egypt's water losses. Wheeler et al. [5] examine cooperative filling strategies with a hydrological model, optimizing release schedules during filling to minimize downstream shortages. Kansara et al. [6] analyze the filling process with Sentinel-1 SAR imagery (2017–2020), finding a contrasting increase in water upstream of the dam and a decrease downstream in August and September 2020. Mordos et al. [7] concur, predicting a 30% decrease in Blue Nile runoff within Sudan during the first filling and drastic long-term changes to the flow regime that render the operating policies of existing Sudanese dams invalid. Elsayed et al. [8] evaluate cooperation under hydrologic uncertainty with a nexus-based approach, showing that cooperation improves basin-wide hydropower and water–food–energy outcomes, particularly for Egypt and Sudan, at a modest cost to GERD's output. Aziz et al. [9] suggest the opposite, reporting potentially severe downstream effects on surface and groundwater levels, saltwater intrusion, and soil salinity.

Most of these studies adopt technically intensive, hydrology- or optimization-heavy methodologies that, while detailed, often omit the evolving socioeconomic and political pressures that influence real-world water management. Many focus on the initial filling period [5]–[7] or evaluate equilibrium states through optimization [4], without explicitly modeling how growing demands and political dynamics interact over time after filling. Our study employs a deliberately simplified, high-level system dynamics framework that captures linearly increasing pressures in the post-filling steady state and integrates the dependencies between water availability, domestic and agricultural demand, energy needs, and diplomatic feedback,

which are often fragmented or externalized in specialized studies. It complements rather than replaces prior models: we do not aim for fine-grained hydrological predictions or exact operating schedules, but for plausible scenarios that reveal emergent trade-offs, leverage points, and the contrast between cooperative and non-cooperative strategies in a qualitative but structured manner.

III. MODEL

We developed the models in Vensim PLE, a system dynamics modeling and simulation platform. Users can explore hydrological and policy scenarios by adjusting release fractions, reservoir capacities, withdrawal fractions, and socio-political pressures through sliders, and observe their impacts on reservoir storage, releases, and interstate interactions.

A. System Structure

The Nile begins with two major tributaries: the White Nile, originating in the Great Lakes region (primarily Lake Victoria, flowing through Uganda and South Sudan), and the Blue Nile, rising from Lake Tana in the Ethiopian Highlands. They converge at Khartoum to form the Main Nile, which flows north through Sudan and Egypt to the Mediterranean. The White Nile contributes a steadier year-round flow, while the Blue Nile brings highly seasonal floodwaters during the Ethiopian rainy season. We model these flows with three key reservoirs: GERD on the Blue Nile in Ethiopia, the Jebel Aulia Reservoir (JAR) on the White Nile in Sudan, and AHD on the Main Nile in Egypt, each governing flow regulation and withdrawals for its country.

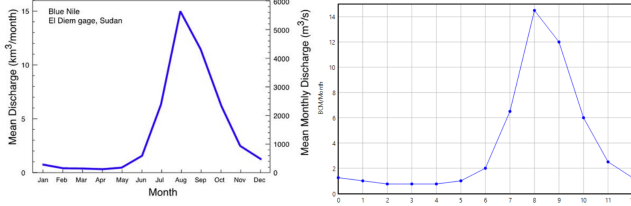
The model uses stock–flow structures on a monthly time step, capturing seasonal fluctuations and longer-term dynamics. Stocks represent reservoir volumes and the reaches of the White, Blue, and Main Nile (BCM); flows represent inflows, releases, evaporation, and withdrawals (BCM/month), completing the journey of the water from its lake sources to the sea. Evaporation is set to 0.001 times each reservoir's volume per month, a simplification chosen for computational efficiency. Dedicated stocks, reset yearly, track the cumulative water available for irrigation and energy generation in each country; these allow calibration against real-world figures such as Egypt's 45 BCM of annual agricultural water use, Sudan's 18.5 BCM allocation, and GERD's projected annual generation of about 15,000 GWh. Table I lists the physical constraints. Overtopping mechanisms release water above maximum capacity, overriding the maximum safe outflow to simulate flooding, which is critical during the high inflows of July–September.

B. Calibration

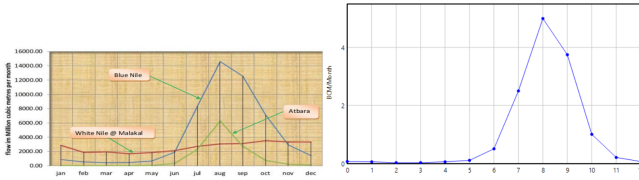
Inflows capture seasonal variability, critical for GERD's filling and the water balance. Blue Nile inflow uses a Vensim LOOKUP function with monthly values from the literature [10] (Fig. 1a). White Nile inflow is modeled as roughly 2 BCM/month with small random additive noise and extra flow during the wet season. Atbara inflow was introduced

TABLE I
RESERVOIR CONSTRAINTS

	GERD	JAR	AHD
Maximum safe capacity (BCM)	74	3.5	162
Minimum safe capacity (BCM)	10	0.5	60
Maximum outflow (BCM/month)	10	3	10
Initial volume, first model (BCM)	5	3	100



(a) Blue Nile: observed [10] (left) and simulated (right)



(b) Nile tributaries: observed [11] (left) and simulated Atbara (right)

Fig. 1. Mean monthly inflows used for calibration (Vensim LOOKUP functions, BCM/month).

during calibration, since AHD storage was otherwise consistently lower than historical levels, and is also modeled as a LOOKUP [11] (Fig. 1b). AHD receives the Main Nile flow after the confluence, net of Ethiopian and Sudanese withdrawals, and its outflow is a LOOKUP based on historical data [12]. Inflows were calibrated to reflect pre-GERD operations and the 1959 allocations (55.5 BCM/year for Egypt, 18.5 BCM/year for Sudan).

C. First Model: Hydrological Framework

The first model establishes a hydrological framework: it simulates GERD's three-stage filling, ensures water balance, and tests operational scenarios under two alternative policy assumptions. It excludes socio-political pressures and relies on fixed release fractions and seasonal LOOKUP patterns; the simulation covers 120 months spanning GERD's filling. Sudanese withdrawals and JAR releases are constant ratios, calibrated to remain within safe volumes while satisfying Sudan's allocation, and GERD follows a three-phase filling policy dependent on its volume, capped at a target volume beyond which inflow is released as is. The two policies mainly differ at AHD:

- *Flow priority* sustains historical outflows (e.g., AHD's 4–6 BCM/month) without adjusting for upstream operations, allowing reservoir volumes to change drastically.
- *Capacity priority* adjusts releases to target specific capacities, reducing releases when storage is low and increasing them near maximum capacity.

Neither policy includes coordinated releases or cross-country feedback, unlike those agreed in the 2015 Declaration of Principles; both therefore represent non-cooperative transboundary policies. Capacity priority, being closer to reality, is carried forward to the scenario simulations, in which switchable controls simulate malfunctions or strategic actions at JAR and GERD (Section IV-B).

D. Second Model: Abstract Socio-Political Pressures

The second model keeps the hydrology and water balance of the first unchanged, assumes a filled GERD, and adds abstract socio-political pressures as dimensionless variables that grow linearly over time, with the potential for future grounding in real-world data. Both flows and capacities are adjustable within the minimum and maximum safe capacities. Six pressures represent internal national demands: energy and agriculture in Ethiopia, and agriculture and domestic needs in Sudan and in Egypt. Each starts at an initial value and increases at a constant monthly rate, reflecting population or development growth; agricultural pressures additionally follow a seasonal LOOKUP pattern peaking in August. The pressures directly influence each country's release and withdrawal fractions through simple additive formulations; for instance, higher Ethiopian energy pressure increases GERD's release fraction, and Sudan's domestic and agricultural pressures drive its withdrawals from the Blue Nile.

Cross-country feedback is intentionally limited to a single diplomatic pressure from Egypt on Ethiopia: when AHD's level falls below a critical threshold, GERD's releases increase, a proxy for negotiations or international pressure. Feasibility flags activate when a country's internal demands cannot be met, signaling unrealistic or unsustainable scenarios during stress testing. Calibration is carried over from the first model; pressure growth rates were chosen to reflect demographic projections, and the model was validated to reproduce AHD releases and GERD's steady-state operation from the first model.

IV. RESULTS

A. Hydrological Model: GERD Filling

The baseline simulations model GERD's filling under both policies, and a comparative setting without GERD (Figs. 2–4). Without GERD, AHD stabilizes at about 100 BCM, sustaining around 45 BCM/year for irrigation and around 10,000 GWh/year of hydroelectric energy. Under flow priority, GERD reaches its target by the third year while AHD storage declines by 30 BCM, to 70 BCM, as reduced inflows sustain irrigation and energy demands. Under capacity priority, AHD maintains 100 BCM, but releases drop by 0.5–1 BCM/month, reducing downstream irrigation by 1.5 BCM/year and energy by 1,500 GWh/year. JAR is unaffected in both cases owing to its independence from the Blue Nile, although Sudan's overall withdrawals are slightly reduced. Continuing pre-GERD policies thus leads to a decline in either the releases or the capacities of downstream reservoirs, underscoring the need for coordinated filling and release policies.

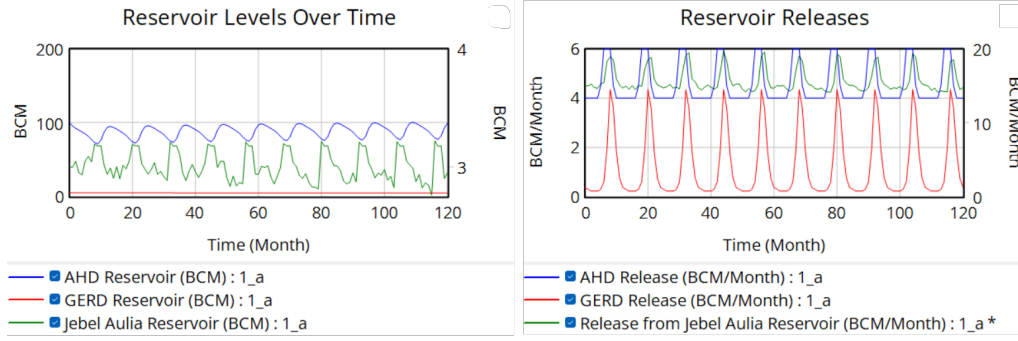


Fig. 2. Reservoir levels and releases without an operational GERD.

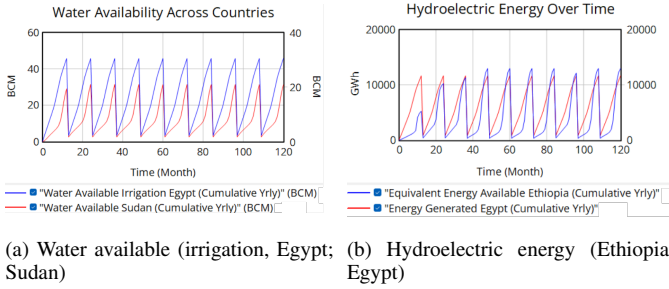


Fig. 3. Baseline yearly-cumulative water availability and hydroelectric energy.

B. Hydrological Scenarios

Jebel Aulia malfunction. To simulate the effect of political instability on operations, JAR's efficiency can either follow time-varying LOOKUP tables reflecting the 2019 Sudanese Revolution and the 2023 conflict between the Sudanese Armed Forces (SAF) and the Rapid Support Forces (RSF), dropping to about 0.65 at the peak of the conflict, or be held at a constant value representing sustained SAF control. With constant efficiency, JAR avoids continual overtopping at 0.85, while lower efficiencies lead to long periods of overtopping. The time-varying case confirms this, with multi-year periods of overtopping when efficiency is lowest, around 2023, effectively rendering the dam useless (Fig. 5). A minimum efficiency of about 0.85 is therefore required, and conflict-induced disruptions significantly increase flood risk, consistent with reported challenges in Sudan during 2023.

GERD malfunction. This scenario tests uncontrolled or deliberate high releases from GERD, e.g., excess releases during floods or a gate malfunction, through an additional outflow, combined with Sudan's flood-responsive withdrawal fraction and Egypt's maximum safe outflow (10 BCM/month). High releases strain downstream reservoirs beyond their buffering capacity, causing flood risk, suboptimal operation, and forced emergency releases, particularly when AHD is near its upper limit of 162 BCM (Fig. 6). Coordinated contingency plans, such as preemptive releases or increased Sudanese diversion, reduce AHD strain; Egypt's infrastructural limits on flood mitigation remain a vulnerability.

C. Pressure-Driven Scenarios

We consider a configuration feasible if no infeasibility flag is triggered for more than two consecutive months in a year, since accommodating wet seasons renders many configurations infeasible under a stricter criterion.

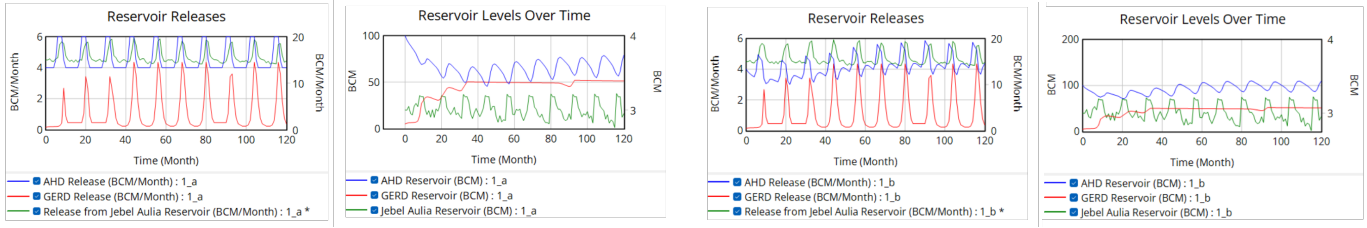
Constant base pressures. With constant pressures reproducing the base conditions of the hydrological model, a feasible solution is obtained with minimal tweaking of flow limits and capacities (Fig. 7), suggesting that constant, moderate pressures can be sustained within current reservoir capacities and practices. Starting pressures had to be tuned to translate into the countries' real agricultural and energy demands; e.g., Egypt's agricultural pressure had to be much lower than unity to not exceed 45 BCM/year of irrigation water, and Sudan's set to 1.5 to achieve about 18 BCM/year.

High constant pressures. Elevated pressures (agricultural pressure in Egypt of 1, domestic pressure in Sudan of 3, and energy pressure in Ethiopia of 2) could not be made sufficiently feasible (Fig. 8). The best configuration found reduces Sudanese withdrawals from the Blue and Main Nile, which prevents prolonged infeasibility for any single country rather than favoring Sudan alone. Annual water availability settles to volumes similar to before despite a substantial decrease in AHD's volume: the inflows themselves impose an upper limit that such pressures exceed. Egypt's demands risk depleting AHD, and Sudan's withdrawals exacerbate downstream shortages.

Linearly growing pressures. Starting from the base values, with Egypt's agricultural, Ethiopia's energy, and Sudan's domestic pressures each growing at 0.01/month, operations become increasingly unsustainable: AHD and JAR volumes deplete and the infeasibility flags, notably Egypt's, activate increasingly often (Fig. 9).

V. DISCUSSION

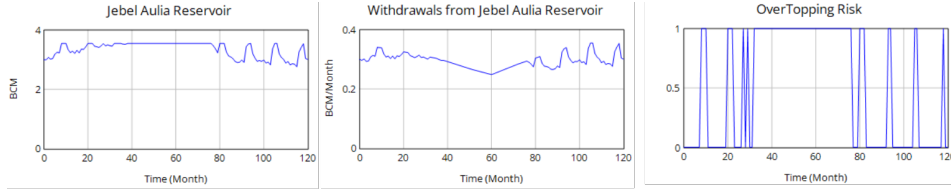
The first model shows that under flow priority, GERD's filling causes significant AHD storage declines that threaten Egypt's water security, whereas capacity priority stabilizes reservoirs at the cost of downstream flows for irrigation and energy. The JAR scenarios underscore Sudan's operational vulnerability, particularly during conflict, with efficiencies



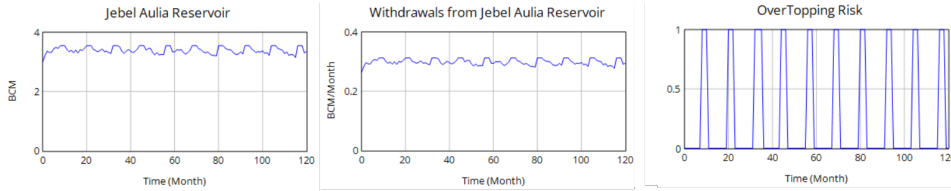
(a) Flow priority

(b) Capacity priority

Fig. 4. Reservoir releases and levels during GERD's filling under the two policies.

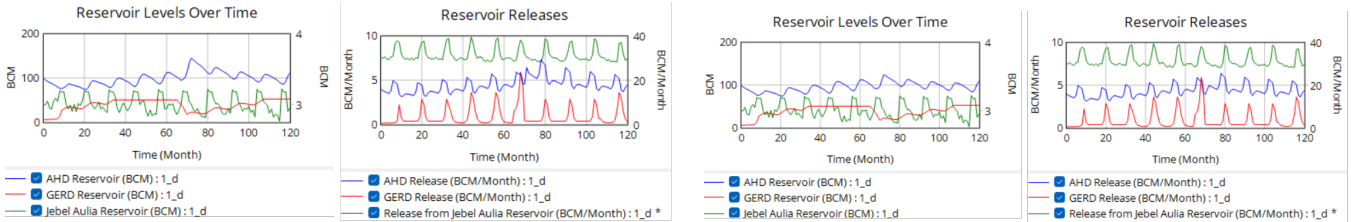


(a) Time-varying efficiency



(b) Constant efficiency of 0.85

Fig. 5. JAR volume, withdrawals, and overtopping risk.



(a) Additional 5 BCM, non-cooperative Sudan: AHD filling and excess releases

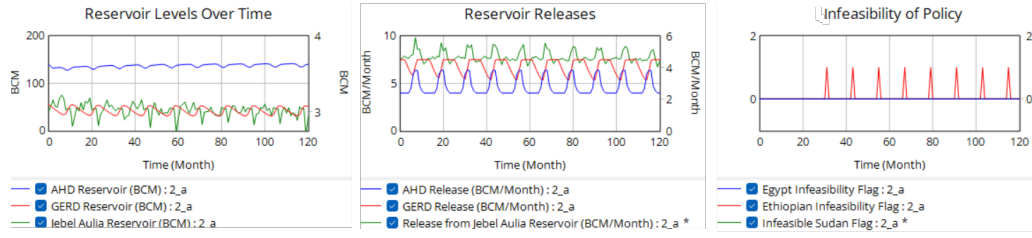
(b) Additional 6 BCM, cooperative Sudanese diversion: sustained AHD operation

Fig. 6. Downstream effects of excess GERD releases.

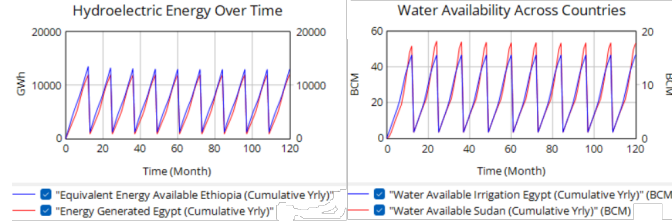
below 0.85 causing frequent overtopping, and the GERD scenarios highlight cascading flood risks, mitigated by Sudanese withdrawals but constrained by Egypt's outflow limits. The second model shows that constant pressures are manageable, whereas high or growing pressures, chiefly agricultural and domestic, drive operations to infeasibility against the physical limit set by the inflows. These findings suggest four policy needs: coordinated filling and release policies that balance stakeholder needs, robust JAR management for Sudan's stability, contingency plans for malfunctions, and negotiated pressure limits to avoid infeasible operation. The Egypt–Ethiopia diplomatic feedback included in the second model is a first step toward representing such coordination.

Limitations and future work. The pressures are abstract

and grow linearly, and cross-country feedback is limited to a single link. Future work could link pressures to tangible metrics (Ethiopia's hydropower output, Sudan's agricultural yields, Egypt's per-capita domestic consumption) using regional economic and hydrological data; expand cross-country feedback, e.g., Sudan's influence on Egypt and Ethiopia, possibly as functions of economic losses or political agreements; systematically evaluate the effect of diplomatic feedback and coordinated releases; and introduce climate variability through stochastic inflows and socioeconomic shocks through dynamic growth rates, to test long-term resilience under drought or population spikes.

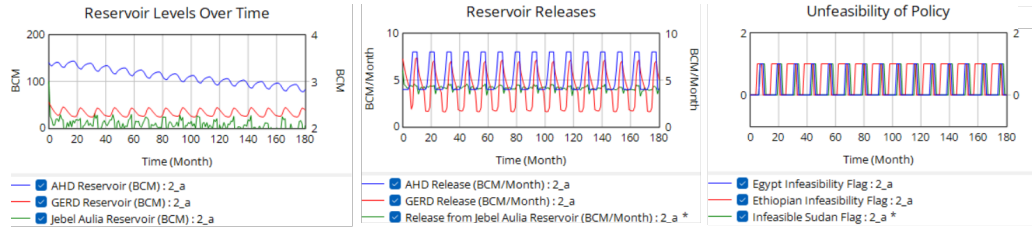


(a) Reservoir levels, releases, and infeasibility flags

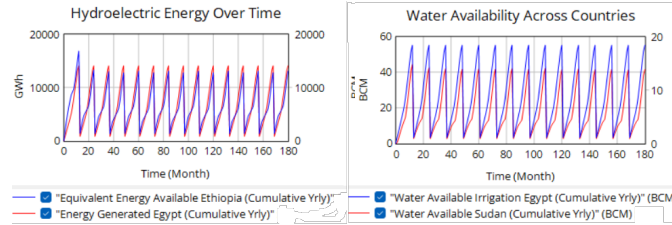


(b) Energy and water availability

Fig. 7. Constant base pressures: feasible operation meeting the required water use.



(a) AHD depletion and infeasibility flags



(b) Energy and water availability

Fig. 8. High constant pressures: required water use is not achieved.

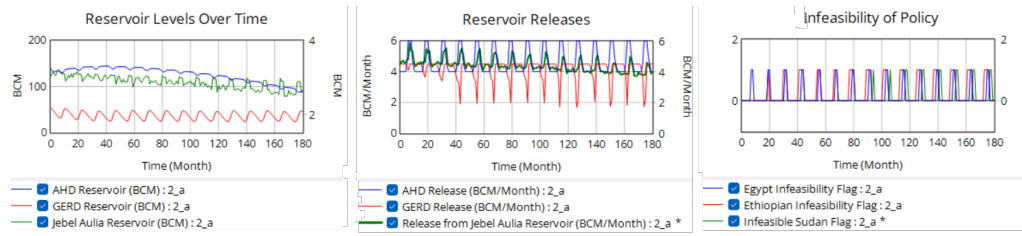
VI. CONCLUSION

This study developed a system dynamics flight simulator in Vensim PLE to analyze water management in the Nile River Basin, simulating GERD, AHD, and JAR across Ethiopia, Sudan, and Egypt. The first model, calibrated with real-world Blue Nile, White Nile, and Atbara inflows and historical allocation agreements, simulated GERD's three-stage filling and tested flow- and capacity-priority operation and malfunction scenarios; the second overlaid abstract, linearly growing socio-political pressures with minimal Egypt–Ethiopia diplomatic feedback. The results underscore the importance of coordinated dam operations that align Ethiopia's energy ambitions with the irrigation and domestic needs of Sudan and Egypt. The model offers a practical framework for scenario analysis

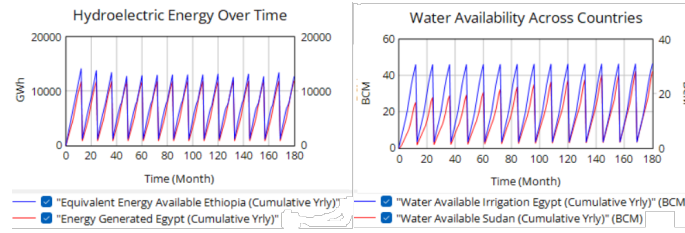
and stakeholder engagement in a geopolitically sensitive region, with potential adaptability to other transboundary river systems facing similar operational and political complexities.

REFERENCES

- [1] UNESCO, *The United Nations World Water Development Report 2020: Water and Climate Change*. Paris: UNESCO, 2020.
- [2] UN-Water, "Transboundary waters: Sharing benefits, sharing responsibilities," United Nations, 2008. [Online]. Available: https://www.unwater.org/sites/default/files/app/uploads/2017/05/UNW_TRANSBOUNDARY.pdf
- [3] Nile Basin Initiative, *State of the River Nile Basin 2020*. Entebbe, Uganda: Nile Basin Initiative, 2020.
- [4] M. Basheer, V. Nechifor, A. Calzadilla *et al.*, "Collaborative management of the Grand Ethiopian Renaissance Dam increases economic benefits and resilience," *Nature Communications*, vol. 12, p. 5622, 2021.



(a) AHD depletion and infeasibility flags



(b) Energy and water availability

Fig. 9. Linearly growing pressures: increasingly infeasible water requirements.

- [5] K. G. Wheeler, M. Basheer, Z. T. Mekonnen, S. O. Eltoum, A. Mersha, G. M. Abdo *et al.*, "Cooperative filling approaches for the Grand Ethiopian Renaissance Dam," *Water International*, vol. 41, no. 4, pp. 611–634, 2016.
- [6] P. Kansara *et al.*, "An assessment of the filling process of the Grand Ethiopian Renaissance Dam and its impact on the downstream countries," *Remote Sensing*, vol. 13, no. 4, p. 711, 2021.
- [7] M. A. Mordos *et al.*, "Hydrological impacts of the Grand Ethiopian Renaissance Dam (GERD) on River Nile hydrology within Sudan," *Hydrology*, vol. 8, no. 3, pp. 41–51, 2020.
- [8] H. Elsayed *et al.*, "Water–food–energy nexus for transboundary cooperation in Eastern Africa," *Water Supply*, vol. 22, no. 4, pp. 3567–3587, 2022.
- [9] S. A. Aziz *et al.*, "Assessing the potential impacts of the Grand Ethiopian Renaissance Dam on water resources and soil salinity in the Nile Delta, Egypt," *Sustainability*, vol. 11, no. 24, p. 7050, 2019.
- [10] D. Conway, "A water balance model of the Upper Blue Nile in Ethiopia," *Hydrological Sciences Journal*, vol. 42, no. 2, pp. 265–286, 1997.
- [11] A. Kidus, "Long-term potential impact of Great Ethiopian Renaissance Dam (GERD) on the downstream eastern Nile High Aswan Dam (HAD)," *Sustainable Water Resources Management*, vol. 5, 2019.
- [12] I. Abd-Elaty, O. Saleh, H. Ghanayem, T. Grischek, and M. Zelenakova, "Assessment of hydrological, geohydraulic and operational conditions at a riverbank filtration site at Embaba, Cairo using flow and transport modeling," *Journal of Hydrology: Regional Studies*, vol. 37, p. 100900, 2021.